Q1

a)

i) each channel seen as a Bernoulli trial with success probability.

$P(\text{busy} \to \text{free}) = \mu \Delta t$

$P(k \text{ channels} \to \text{idle}) =$

$\binom{i}{k} (\mu \Delta t)^k (1 - \mu \Delta t)^{i-k}$

$P(1 \text{ channel} \to \text{idle}) =$

$\binom{i}{1} (\mu \Delta t)(1 - \mu \Delta t)^{i-1} = i\mu \Delta t + o(\Delta t)$

$\mu_i = i\mu$ [4]

ii) $\lambda_i = (1 - i/M)\lambda_0$

when $M \gg N$ the arrival rate in state i is $\lambda_i = (1 - i/M)\lambda_0$

$\approx \lambda_0$ for all i [3]

iii) using Erlang formula chart

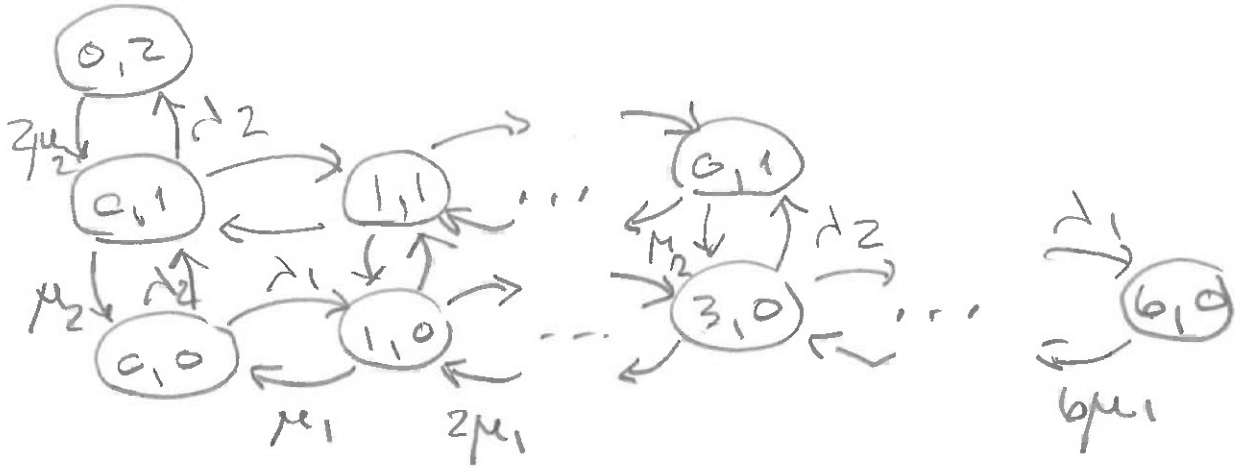
$N \sim 22$

[3]

Q1

b) N_t^1 = Traffic A

N_t^2 = Traffic B



Local Balance equations [5]

$$\pi_i^1 = \left(\frac{\lambda_1}{i\mu_1}\right)\pi_{i-1}^1 = \left(\frac{\rho_1}{i}\right)\pi_{i-1}^1 \quad \text{with } \rho_1 = \frac{\lambda_1}{\mu_1}$$

$$\pi_j^2 = \left(\frac{\lambda_2}{j\mu_2}\right)\pi_{j-1}^2 = \left(\frac{\rho_2}{j}\right)\pi_{j-1}^2 \quad \text{with } \rho_2 = \frac{\lambda_2}{\mu_2}$$

[3]

Recursive

$$\pi_i^1 = \left(\frac{\rho_1^i}{i!}\right)\pi_0^1 \quad \text{and} \quad \pi_j^2 = \left(\frac{\rho_2^j}{j!}\right)\pi_0^2$$

$$\Rightarrow \quad \pi_{ij} = c\left(\frac{\rho_1^i}{i!}\right)\left(\frac{\rho_2^j}{j!}\right) \qquad c = \pi_0^1\pi_0^2$$

[2]

Q2

a) offered traffic $3 \times 3 = 9$ erlangs

mean $M = \rho_0 E_{15}(\rho_0) = 0.18\ m$

variance $V = m\left(1 - M + \frac{\rho_0}{M + m + 1 - \rho_0}\right)$

$= 0.373$

$(M = 1.5)$

[2]

$E(N_t) = \frac{\alpha}{\alpha + \beta} \rho_0 = 0.18$

[2]

$E(N_t(N_t - 1)) = \left(\frac{\alpha}{\alpha + \beta}\right)\left(\frac{\alpha + 1}{\alpha + \beta + 1}\right) \rho_0^2$

$= Var[N_t] + [E(N_t)]^2 - E[N_t]$

$= 0.2254$

[2]

$\alpha = 0.145$

$\beta = 7.108$

$\lambda = 3$

$\mu = 1/3$

[2]

Q2 b)

i) $P(\text{delay}) = P(\text{all } K \text{ servers busy} \mid \text{buffer not full})$

$= P(K \le N_t < K+B)$

$\pi_i = \underbrace{\left(\frac{A^K}{K!}\right)\pi_0}_{\pi_K} \rho^{i-K} \qquad K \le i < K+B$

$j = i-K$

$0 \le j < B$

$$P(\text{delay}) = \pi_K \sum_{j=0}^{B-1} \rho^j \qquad [2]$$

$$= \pi_K \left(\sum_{j=0}^{\infty} \rho^j - \sum_{k=B}^{\infty} \rho^k \right)$$

$$= \pi_K \left(\sum_{j=0}^{\infty} \rho^j - \rho^B \sum_{k=0}^{\infty} \rho^k \right)$$

$$= \pi_K \left(\frac{1}{1-\rho} - \frac{\rho^B}{1-\rho} \right)$$

$$= \pi_K \left(\frac{1-\rho^B}{1-\rho} \right) \qquad [3]$$

$P_{loss} = P(\text{buffer full}) = P(N_t = K+B)$

$= \pi_K \rho^B$

$\pi_K = \frac{A^K}{K!}\pi_0$ [3]

Q2 b) $M/M/2/5$

$\lambda = 120 \quad \mu = 100 \quad B = 3$

$A = \frac{120}{100} = 1.2 \qquad \rho = \frac{120}{200} = 0.6$

$E_K(A) = E_2(1.2) = 0.25$

$\pi_K = \frac{1}{5.226}$

$P[\text{loss}] = \pi_K \rho^B$

$= \frac{1}{5.226}(0.6)^3$ [2]

$=$

$P[\text{delayed}] = \pi_K \left(\frac{1-\rho^B}{1-\rho} \right)$

$= \frac{1}{5.226} \left(\frac{1-(0.6)^3}{1-0.6} \right)$

$=$ [2]

3a) M/G/1

mean message length $= \frac{1}{3}B + \frac{1}{4}2B + \frac{5}{12}3B$

$$E(S) = \left(\frac{4}{12} + \frac{6}{12} + \frac{15}{12}\right)B = \frac{25}{12}B = 2.08 \cdot 160 \text{ bits}$$

$$= 333.33 \text{ bits}$$

$$= 5.20 \text{ ms}$$

[2]

mean square message length =

$$E(S^2) = \frac{1}{3}B^2 + \frac{1}{4}(2B)^2 + \frac{5}{12}(3B)^2 =$$

[2]

$$\left(\frac{1}{3} + \frac{4}{4} + \frac{45}{12}\right)B^2 = \left(\frac{4+3+45}{12}\right)B^2$$

$$= 4.33\,B^2 = 4.33 \cdot \frac{(160)^2}{(64)^2} = 27.06 \text{ (ms)}^2$$

$$E(W) = \frac{\lambda E(S^2)}{2(1-\rho)}$$

[2]

$$\rho = \lambda E(S) = 150 \times \frac{333.33}{64000} = 0.78$$

$$E(W) = \frac{150 \times 27.06}{2(1-0.78)} = 9.225 \text{ ms}$$

[1]

$$E(T) = E(W) + E(S) = 14.425 \text{ ms}$$

[2]

3b)

i) large number of sources

$N \gg 1 \quad P \ll 1$

$$\pi_i = \binom{N}{i} P^i (1-P)^{N-i}$$

can be approximated by a normal distribution

and $$P_L = \sum_{i=j_0}^{N} \frac{(i-c)\pi_i}{m} \sim \frac{1-P}{c-m}\varepsilon$$

$$\varepsilon = \sum_{i=j_0}^{N} \pi_i \sim \frac{\sigma e^{-\frac{(c-m)^2}{2\sigma^2}}}{\sqrt{2\pi}(c-m)}$$

$$C_{LS} \sim mR_p + \sigma\sqrt{-\ln(2\pi) - 2\ln(\varepsilon)}\,R_p \quad [4]$$

ii) effect of Buffer $\quad G(x) \sim A_N \rho^N e^{-\frac{\beta R_x}{R_p}}$

$$R = \frac{(1-\rho)\left(1+\frac{\alpha}{\beta}\right)}{1-\frac{C_L}{NR_p}} \qquad \rho = \frac{N_p R_p}{C_L}$$

$$C_{LF} = R_p N\left(\frac{1-k}{2}\right) + R_p N\sqrt{\frac{(1-k)^2}{2} + k\rho}$$

[4]

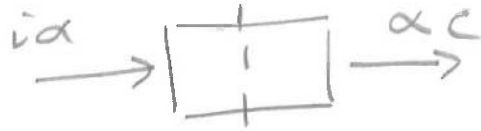
iii) one exaple : call admission control

[3]

4a)

first discuss the following mapping into fluid flow

i sources on — iV cells/s → $\frac{iV}{\frac{V}{\Delta}} = \alpha i$



$\Delta x = (i-c)\alpha \Delta t$ $\quad \frac{\Delta x}{\Delta t} = (i-c)\alpha$

Build up the following set of equations

$$F_i(t+\Delta t, x) = [N-(i-1)]\lambda \Delta t F_{i-1}(t,x) + (i+1)\alpha \Delta t F_{i+1}(t,x) + \{1-[(N-i)\lambda + i\alpha]\Delta t\} F_i[t, \underbrace{(i-c)\alpha \Delta t}_{\Delta x}]$$

...

[3]

$$\frac{\partial F_i(t,x)}{\partial t} = h_1 F_{i-1}(t,x) + h_2 F_{i+1}(t,x) + h_3 F_i(t,x) - (i-c)\alpha \frac{\partial F_i(t,x)}{\partial x}$$

$= 0$ for statistical equilibrium

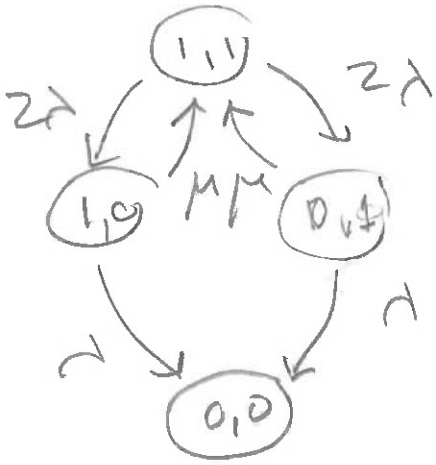
[2]

4b)

i) (1,1) (0,1) (1,0) (0,0) Faulty states

[2]

ii)

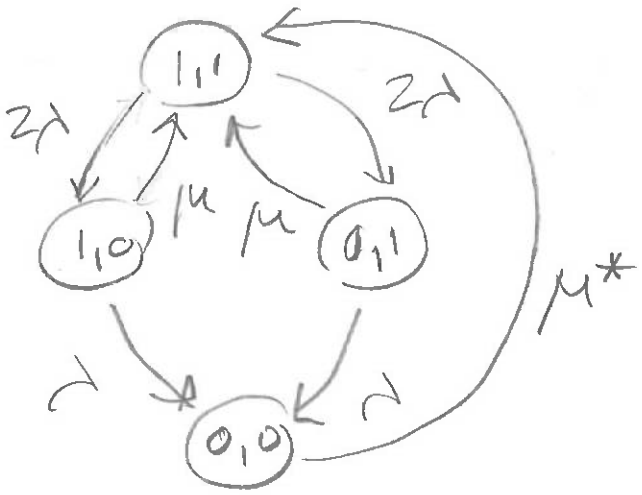


λ = rate of failure

μ = rate of repair

[2]

iii)



[3]

iv) Discussion consider the tradeoff of

- having standby redundant HW onsite with periodic maintenance period visit
- Reactive maintence: just replace HW when it is faulty by a non scheduled visit (emergency)

[3]

9/9

4a)

ii) Use $N=1$

$$M = \begin{bmatrix} -\lambda & \lambda \\ \alpha & -\alpha \end{bmatrix} \qquad \gamma = \lambda/\alpha$$

$$M' = \begin{bmatrix} \frac{\gamma}{c} & \frac{\gamma}{1-c} \\ \frac{-1}{c} & \frac{-1}{1-c} \end{bmatrix}$$

$$zI - M' = 0$$ [2]

Set: $z = 0$

$$z = \frac{\gamma}{c} - \frac{1}{1-c}$$

$$z < 0$$

$$\frac{\gamma}{1+\gamma} < c < 1$$

$$\rho = \left(\frac{\gamma}{1+\gamma}\right)\frac{1}{c} < 1$$

$$z = -\frac{(1-\rho)(1+\gamma)}{(1-c)}$$ [3]